

Computational design and candidate prioritization of putatively pH-sensitive EGFR binders

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Computational manuscript; evidence reviewed through 7 October 2026.

Abstract

We designed small proteins intended to bind the epidermal growth factor receptor (EGFR) more strongly at pH 6.5 than at pH 7.4, while retaining recognition of mouse EGFR at pH 6.5. We explored two mechanisms. Designed domain 1 (DD1) was intended to bind EGFR through contacts that become more favorable at lower pH. Designed domain 2 (DD2) was intended to cover DD1 at neutral pH and uncover it under acidic conditions. Four hundred RFdiffusion3 designs yielded two starting binders that passed structural filters for both species. Proton-PottsMPNN generated 3,000 sequence variants from three previously selected DD1 binders. Among 30 variants selected by predicted interface confidence (ipSAE), three—062/0955, 062/0629 and 062/0436, named by starting binder/variant—had an acid-favored model score in both species when the estimated ionization midpoints (pKa values) were shifted by -1 , 0 and $+1$. This finding held for the original pH 6.5/7.3 analysis and for its pH 6.5/7.4 Henderson–Hasselbalch reweighting: only the probabilities assigned to existing protonation-state scores were changed; no new Potts evaluation at pH 7.4 was performed. We also generated 1,000 DD2 extensions. The assessed 91-residue fusion, D46, preserved the DD1 fold. Further sampled structures placed DD2 near or clear of the reference binding face, providing geometric examples for assessing the masking hypothesis. AlphaProtein Novo and LigandForge each supplied a further 1,000 starting designs; Proton-PottsMPNN generated 10,000 redesigns from each platform's ten selected parents. One hundred redesigns per platform, selected by paired interface confidence, were assessed for affinity and receptor-histidine pKas. These results identify candidates for experimental testing and a structural requirement for future masks. They do not establish pH-dependent binding or switching experimentally.

Keywords: EGFR; protein design; histidine protonation; pH sensitivity; intramolecular masking.

1 Introduction

Our goal was to design a binder that recognizes human EGFR more strongly at pH 6.5 than at pH 7.4 and also recognizes mouse EGFR at pH 6.5. We considered two ways to achieve this. The first changes the contacts between the binder and EGFR. The second changes whether the binding surface is exposed. Designed domain 1 (DD1) is the binding part of the protein; designed domain 2 (DD2) is an attached extension intended to cover DD1 at neutral pH. At lower pH, DD2 was intended to become less stable or unfold, exposing DD1 (Fig. 1).

We use *complex* for a binder and receptor modeled together. The two proposed effects can be expressed separately:

$$K_{a,\text{app}}(\text{pH}) = f_{\text{accessible}}(\text{pH}) K_{a,\text{intrinsic}}(\text{pH}). \quad (1)$$

Here, $K_{a,\text{intrinsic}}$ describes how strongly exposed DD1 binds EGFR. The fraction $f_{\text{accessible}}$ is the proportion of unbound fusion molecules with an exposed DD1 surface, between zero and one. Their product, $K_{a,\text{app}}$, describes binding by the whole fusion. Lower pH could strengthen binding by improving the contacts, exposing more DD1 molecules, or both. This model assumes that the mask can equilibrate rapidly and that EGFR binds only exposed DD1. Neither factor was measured in this study.

The chemical basis is protonation: a chemical group gains a proton, which can change its charge. Histidine can become positively charged as pH falls, whereas aspartate can provide a negatively charged carboxylate group. We therefore placed aspartates in DD1 near two EGFR histidines, His37 and His100, to favor contact under acidic conditions. The intended electrostatic effect is to stabilize the protonated histidines, potentially increasing their protonation probability; this remains a design hypothesis. These positions refer to the 192-residue Domain III (D3) fragment used for design, rather than the full receptor sequence. A group's pKa is the pH at which its protonated and unprotonated forms are equally probable in a simple titration model. Its surroundings can shift that value, so pKa prediction is relevant to the intended 6.5–7.4 window.

Protein thermodynamics provides the basis for linking protonation to binding [1, 2], and earlier designs used histidine contacts to control protein shape and assembly [3, 4, 5]. Ahn and colleagues designed binders that release their targets under acidic conditions by weakening the interface or destabilizing the binder [6]. Our masking proposal uses weakening within the fusion to expose the binding surface.

Proton-PottsMPNN is an existing sequence-design method that distinguishes charged and neutral forms of ionizable residues. Jacobsen and colleagues used it to recover PD-L1 binders that engage under acidic conditions, with reported transition pHs of 4.0–5.8 [15]. They selected the redesigned experimental library using the model’s protonation-aware objectives; the starting designs, but not the redesigned library, had been filtered by structure prediction. Our workflow instead kept each starting protein backbone fixed, ranked new sequences by predicted interface confidence, and then evaluated a separate pH-dependent model score [14, 15, 18]. The distinctive questions here are whether this workflow identifies candidates for the near-neutral 6.5–7.4 window, retains recognition of both EGFR species, and can produce a working attached mask.

Most calculations used EGFR D3. The intended experiments use the full extracellular part of EGFR, whose motion, sugar modifications (glycans) and neighboring domains can affect access to the binding surface [27]. We therefore assess human and mouse models separately. The results identify three DD1 variants for follow-up and show that D46 retained the binding fold and sampled different DD2 placements without establishing a functional mask. AlphaProtein Novo and LigandForge provide two additional design routes, each assessed through 100 selected Proton-PottsMPNN redesigns with paired confidence, affinity and receptor-histidine pKa predictions [16, 17, 14, 15, 18, 20].

For readability, the starting DD1 binders are called DD1-062, DD1-195 and DD1-310. A variant such as 062/0955 is sequence 0955 redesigned from DD1-062. The assessed fusion is called D46, and AlphaProtein Novo designs use names such as Novo-0293. Supplementary section S8 maps these short names to the exact saved identifiers; the sequence records themselves have not been renamed.

2 Results

2.1 Design campaigns addressed binding contacts and masking separately

The first design aim combined a plausible EGFR-binding surface with contacts that could respond to pH. The second aimed to control access to DD1 through attached DD2 (Fig. 1). Table 1 lists the campaigns and their sizes; these counts overlap and cannot be added to obtain independent binders. Figure 3 compares the same 394 candidate sequences in all five panels. All have paired confidence, affinity and Potts scores. Receptor histidine pKas are unforced for 392 candidates; the two remaining mouse His100 cases are displayed under both explicit neighboring-acid assumptions. The source-linked inventory is retained in Supplementary section S21. A separate DD2-as-a-second-chain experiment is described in Supplementary section S4; it did not test the attached-mask mechanism.

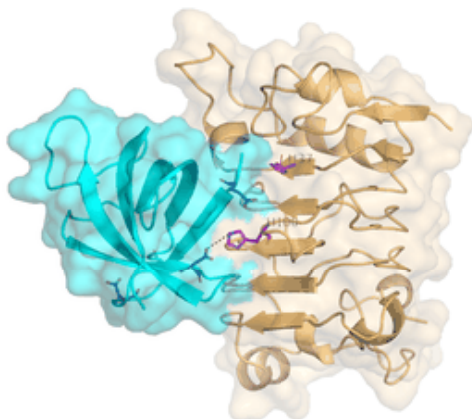
2.2 A pH-dependent model score prioritizes three redesigns for testing

RFdiffusion3 placed two aspartates near EGFR D3 His37/His100 and generated 400 DD1 designs. Two starting binders passed both interface-confidence (ipSAE > 0.63) and fold-similarity (TM-score > 0.75) filters for human and mouse EGFR. This campaign was separate from the three previously selected DD1 binders used for Proton-PottsMPNN redesign. That redesign produced 1,000 variants per starting binder. Selecting ten per starting binder by mean human/mouse ipSAE gave 30 sequences and 60 predicted complexes. All 120 EGFR histidine pKa calculations without imposed neighboring-acid states and all 60 DeltaForge requests returned numerical results.

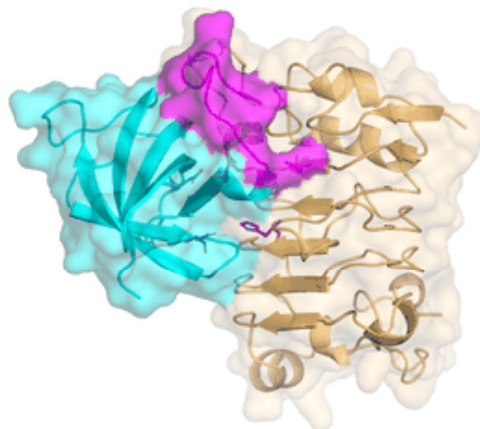
Figure 3B asks which of these 30 confidence-selected variants also have a model score that improves under acidic conditions in both species. Lower Potts scores are more favorable in this model [15]. This differs from ipSAE, where higher values indicate greater interface confidence; the two scores measure different things. For each predicted complex, we used the saved scores for different combinations of protonated DD1 residues. Each score compares the complex with its two separate protein partners. PROPKA pKa estimates determined the probability

A · D46 DD1 only + human EGFR

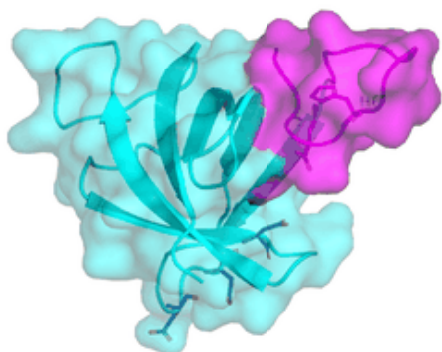
Boltz-2; matched parent without the DD2 extension

**B · D46 fusion + human EGFR**

Boltz-2; native complex prediction

**C · Intended pH 6.5: DD1 exposed**

DD2 clear of binding face; cluster 3, seed 195 / sample 3

**D · Intended pH 7.4: DD1 covered**

DD2 near binding face; cluster 4, seed 149 / sample 2

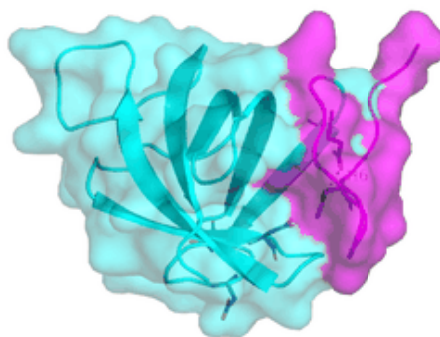


Figure 1: D46's matched binding module and DD2–DD1 fusion. (A) The 77-residue DD1-only parent in a retained Boltz-2 prediction with human EGFR D3. Its sequence is identical to residues 15–91 of the D46 fusion. (B) The intact 91-residue fusion in its retained Boltz-2 target-complex prediction [8]. (C) Intended acidic-pH form (pH 6.5): DD2 is displaced from DD1's reference binding face, leaving DD1 exposed (cluster 3, seed 195/sample 3). (D) Intended neutral-pH form (pH 7.4): DD2 covers part of that face (cluster 4, seed 149/sample 2). These AFSample3 examples illustrate the proposed design [11, 12]. Only the fusion chain is displayed in C/D. Cyan marks DD1, magenta DD2 and light orange human EGFR; highlighted histidine and carboxylate carbons are purple and blue, respectively, with standard nitrogen and oxygen colors. Semi-transparent surfaces reveal the cartoons. The dashed line in A marks the 3.63 Å Asp46–His100 separation; that in D marks the 2.92 Å DD2 His6–DD1 Glu16 separation. These are distances, not verified hydrogen bonds. The pH labels denote the intended design forms. The sampled structures did not use pH as an input and do not establish unfolding or a pH-dependent transition; they are selected examples, not cluster medoids. The archived UMAP projection, all four cluster medoids and the example-selection criteria are provided in Supplementary Figures S1–S2, Table S7 and section S10.

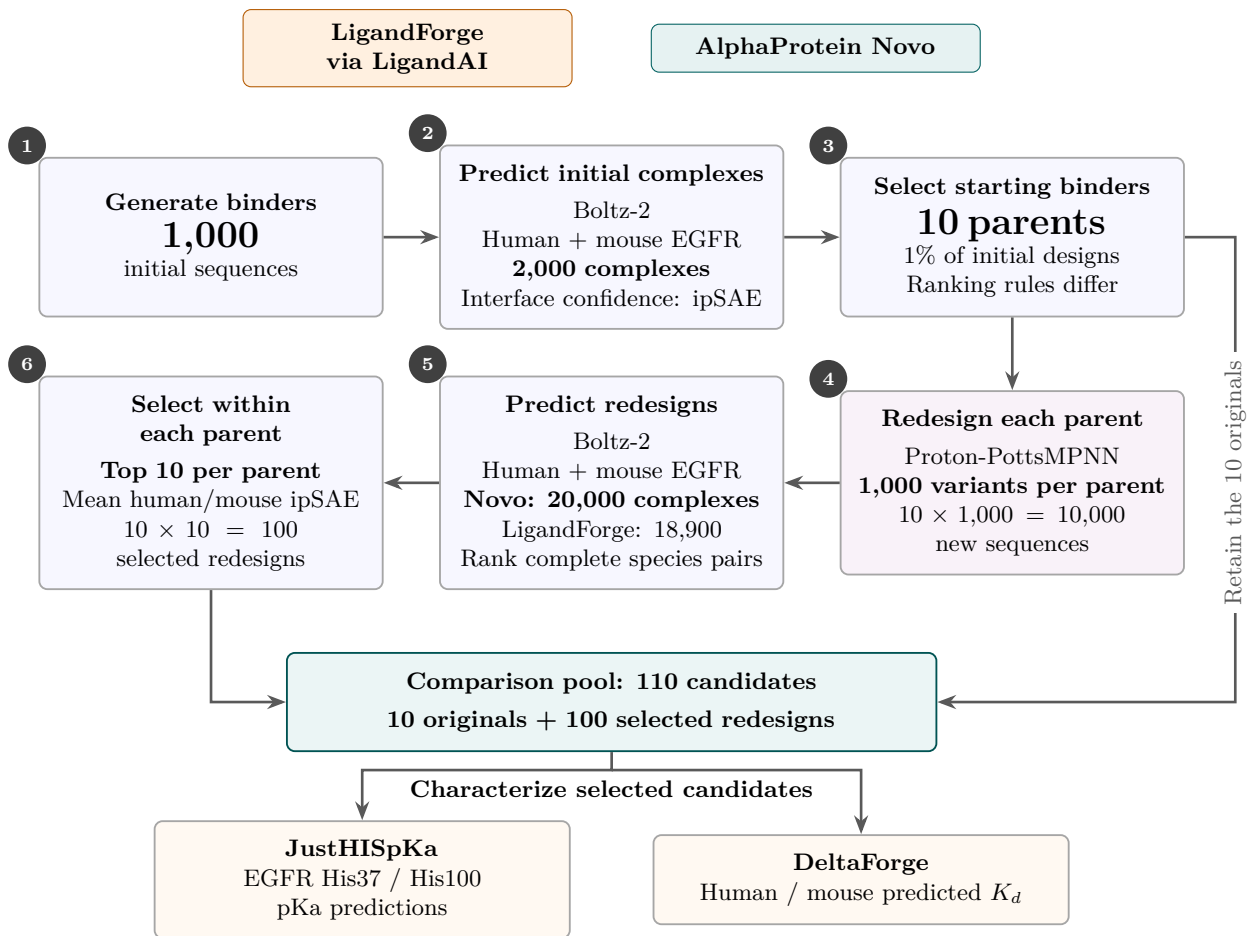


Figure 2: Generation, sequence redesign and candidate assessment for LigandForge through LigandAI and AlphaProtein Novo [17, 16]. The workflow runs independently for each platform; counts are sequences unless labeled as complexes. Each generator supplied 1,000 initial designs, assessed with Boltz-2 against human and mouse EGFR D3 [8]. Ten starting sequences (1% of generated designs) were retained. AlphaProtein Novo ranked mean human/mouse ipSAE; LigandForge ranked species-specific complexes together, yielding nine human-selected and one mouse-selected parent [9]. Standalone Proton-PottsMPNN generated 1,000 fixed-backbone variants per parent, or 10,000 redesigns per platform [14, 15, 18]. Ten redesigns per parent were selected by mean human/mouse ipSAE from complete paired predictions, giving 100 selected redesigns and 110 candidates including the originals for each platform. LigandForge contributes 9,450 completed pairs: nine parents have 1,000 pairs and parent 774 has 450. AlphaProtein Novo contributes all 10,000 pairs, 1,000 per parent. Each selected redesign arm has 400 completed JustHISpKa receptor-site predictions and 200 completed DeltaForge affinity/free-energy predictions [20, 17]. These follow-up scores did not determine ipSAE selection. All 200 redesigns contribute to every panel of Figure 3. Supplementary Figures S3–S4 show initial distributions, S5 shows LigandForge redesign distributions and S6 shows initial Novo and all ten complete redesign libraries. RosettaCommons Foundry is related-framework attribution for the standalone Proton-PottsMPNN implementation. Predicted scores do not establish experimental binding or pH selectivity.

Table 1: Design campaigns, numbers assessed and selection rules. ipSAE estimates confidence in a binding interface; pTM estimates confidence in the whole protein fold. These are prediction-confidence scores, not binding affinities.

Campaign	Numbers assessed	Assessment and selection
Foundry DD1 discovery audit	10,742 unique sequences; 21,484 native human/mouse complexes	Fixed top 100 by mean species ipSAE (0.9309%); three manually selected redesign parents retained separately. The audit overlaps original discovery rows and must not be added to them.
Two-Asp DD1	400 binders; 800 human/mouse complexes	Two designs from placement pair 11 passed interface-confidence and fold-similarity filters in both species.
Covalent DD2 extension	1,000 fusions; 3,000 Boltz-2 calculations	Isolated fusion and human/mouse complexes; five highest-pTM fusions advanced to AFSample3 submission.
Separate-chain DD2	1,040 analyzed complexes per DD1 parent	DD1-195 or DD1-310 with DD2; top 5% selected within each target; no EGFR in these predictions.
AlphaProtein Novo DD1	1,000 designs; 2,000 human/mouse complexes	Ten sequences selected by mean human/mouse ipSAE.
LigandForge DD1	1,000 sequences; 2,000 human/mouse complexes	Ten sequences selected by ranking all human and mouse complexes together; the other species assessed for each sequence.
Proton-PottsMPNN	1,000 variants per parent; 6,000 human/mouse complexes	Ten per parent selected by mean species ipSAE; pH-dependent model scores assessed separately.
Expert DD1 mutants	22 sequences; 220 sampled models	Five samples for each of 44 human/mouse complexes; one model selected per complex.
Novo-derived Proton-PottsMPNN	10,000 variants; 20,000 human/mouse complexes	Ten redesigns per parent selected by mean species ipSAE from ten complete 1,000-pair libraries; 100 selected sequences fully scored.
LigandForge-derived Proton-PottsMPNN	10,000 variants; 18,900 human/mouse complexes	Ten redesigns per parent selected by mean species ipSAE from 9,450 complete pairs; 100 selected sequences fully scored.
Additional rational variants	17 new sequences; 60 sampled models including WT control	Fifteen Novo single substitutions yielded 30 complexes. D46 WT, V8G and V8S each had five samples per species; one model selected per complex.
D46 fusion AFSample3	200 seeds $\times$ five samples	1,000 models for one fusion-human EGFR construct; geometric clustering.

of each combination at a given pH, and we averaged the scores using those probabilities. This calculation assumes that the sites titrate independently. Protein coordinates and all other protonation states were kept fixed.

For the original 30-variant analysis, the pH 6.5/7.4 differences are Henderson–Hasselbalch reweights of saved protonation-state scores from the pH 6.5/7.3 archive, rather than a new Potts evaluation at pH 7.4. In practical terms, we changed the probability weights, not the scores for the individual protonation patterns. Supplementary Table S5 gives both endpoint comparisons for every row of Supplementary Table S6. This calculation changes selected DD1 sites; it does not directly titrate EGFR His37/His100 or model DD2 unfolding.

The plotted quantity is an *acid-favoring Potts difference*: the average score at pH 7.4 minus that at pH 6.5. Lower Potts scores are favorable, so positive values match the intended acidic preference; negative values favor the neutral condition. For example, scores of 10 at pH 7.4 and 8 at pH 6.5 give $10 - 8 = +2$. This is a model preference, not a predicted change in K_d . Figure 3B displays one point for each of the same 394 candidate sequences as the other main panels, with human results horizontally and mouse results vertically. The shaded upper-right region marks acid-favored scores in both species. The supplementary CSVs flag candidates retaining this direction under all three common pKa shifts. The expanded assessment recomputed protonation-pattern scores on the retained native complexes; the earlier 30-variant analysis and its endpoint reweighting remain separately documented in Supplementary Tables S5–S6.

In the original DD1 redesign subset, five of 30 variants met this criterion using the original pKa estimates: four from DD1-062 and one from DD1-310. We then tested sensitivity to the pKa assumptions by shifting all varied-site pKas together by -1 , 0 and $+1$. Three variants—062/0955, 062/0629 and 062/0436—still favored pH 6.5 in both species under every shift (Supplementary Table S6). They ranked 1, 8 and 10 by structural confidence within the DD1-062 group. The numbers meeting the criterion at the three shifts were 5, 5 and 4, but membership changed. This test checks one specified source of uncertainty; it is not a statistical significance test or a complete estimate of prediction error.

The original 30-variant screen therefore identified three sequences to test alongside the structural ranking. Variant 195/0591 had higher mean ipSAE than 062/0955 but favored pH 7.4 in both species according to the Potts score. Confidence in a predicted interface and the desired pH trend select different candidates. The score difference cannot be converted into binding affinity, binding energy or a fold-change in binding. In the 394-candidate main-figure cohort, 104 candidates favored pH 6.5 in both species at the nominal pKas, and 96 retained that direction under all three common shifts. Supplementary Figure S7 and Supplementary Data 4 retain every candidate and both species scores. These counts refer to the selected candidate inventory, not all generated designs. These counts use the binder-site rules described in Methods; sites chosen by proximity for designs outside the Proton-PottsMPNN runs were not necessarily direct interface contacts. A candidate outside the shaded region has not been shown experimentally to lack pH sensitivity. The intended preference must be established by measuring binding at both pHs.

2.3 D46 retained the binding fold and sampled different DD2 placements

For future masks, we propose a structural acceptance rule: a predicted model must show polar contacts between the DD2 histidine and DD1, and DD2 must cover the EGFR-binding surface of DD1 when compared with its receptor-bound structure. If a Lys–His–Lys arrangement is part of the design, those residues must be close in space. Their order in the sequence and a high confidence score for the isolated fusion do not establish masking. This is a proposed rule for assessing future designs; it was not applied during the original generation campaign.

D46 contained the 14-residue DD2 extension **AAAKEHTVGEPLAK** followed by an unchanged 77-residue DD1 (Supplementary section S2). The only histidine was DD2 H6. Lysines K4 and K14 occurred on either side of it in the sequence, but their terminal nitrogen atoms were approximately 10.5/14.6 Å from the nearest histidine ring nitrogen in the RFD3 model, and 10.4/16.0 Å after Boltz-2 structure prediction. These distances did not support the intended direct histidine–lysine contacts in either isolated-fusion model.

The extension nevertheless preserved the DD1 fold. Relative to the original design, DD1 had a TM-score of 0.93895 and a root-mean-square deviation (RMSD) of 1.03 Å. TM-score describes overall structural similarity, with higher values indicating closer agreement; RMSD describes the typical positional difference after alignment, with smaller values indicating closer agreement. For the whole fusion, TM-score was 0.84482 and RMSD 2.87 Å; its predicted fold-confidence score (pTM) was 0.840910. D46 therefore retained the binding module, while the

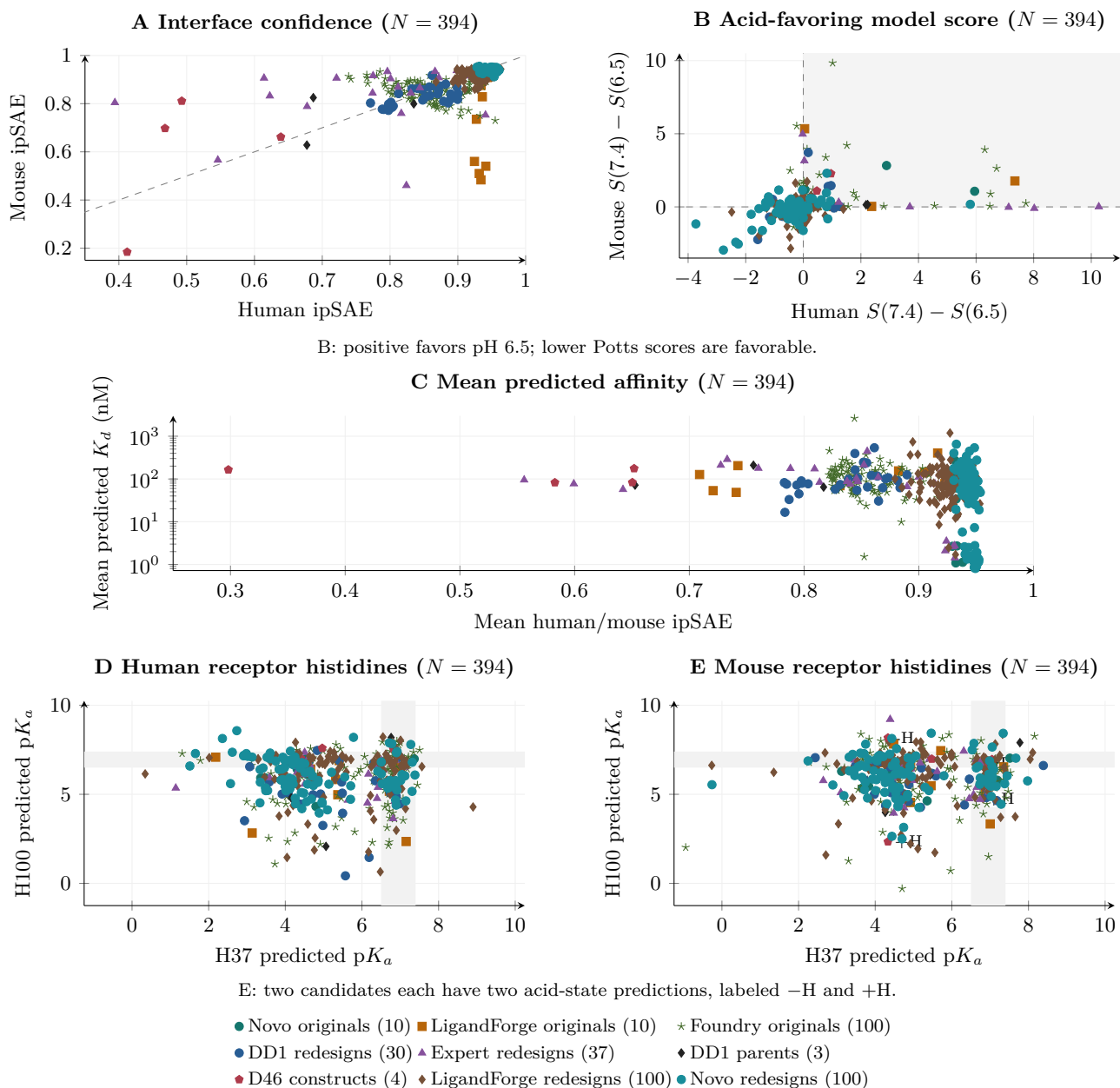


Figure 3: Five assessments of the same 394 candidate sequences, including all 100 Novo-derived redesigns selected as the ten highest mean human/mouse ipSAE values within each complete 1,000-pair parent library. Colors and symbols identify the same cohorts in every panel [7, 14, 15, 16, 17, 18]. (A) Human and mouse interface confidence (ipSAE). (B) Weighted Potts score at pH 7.4 minus that at 6.5; positive values favor the acidic condition because lower scores are favorable. 104 candidates favor acid in both species, and 96 retain this direction under all three common binder-site pKa shifts. Supplementary Figure S7 retains the full score inventory. (C) Arithmetic mean human/mouse ipSAE versus mean DeltaForge predicted K_d [17]; Supplementary Figure S8 separates species. (D,E) EGFR D3 His37/His100 pKas, with the inclusive 6.5–7.4 window shaded. Panel E has 392 unforced pairs plus two predictions each for DD1-062 A76Q+I88Q and the D46 fusion, representing 394 sequences in 396 plotted points. Labels $-H/+H$ mean nearby binder acids were fixed as deprotonated/protonated: Asp55 for the mutant, Glu16/Asp60 for D46. These are separate conditional predictions, not uncertainty bounds or assigned pH states, and are not averaged. Receptor pKas use JustHISpKa [20]; prior PROPKA and receptor-only estimates are preserved separately. Supplementary Figures S9–S10 show the full confidence and receptor-pKa inventory. Data 3A–3E identify every output and assumption. Predicted affinity, receptor ionization and independent-site model scores do not establish measured pH selectivity.

originally reviewed isolated-fusion models did not show the intended histidine-lysine arrangement. Figure 1A,B compares the matched DD1-only and fusion target-complex predictions; Supplementary section S4 gives their scores.

AFSample3 generated 1,000 models that were grouped into four geometric clusters. A subsequent review of DD2 placement supplied the examples in Figure 1C,D. Panel C illustrates the intended pH 6.5 form: none of the sixteen reference binding-face residues approaches DD2, and the closest DD2 atom is 7.31 Å from the receptor placement obtained by aligning DD1 to its matched bound parent. Panel D illustrates the intended pH 7.4 form: DD2 approaches eight of those sixteen DD1 residues, and its His6 ring nitrogen lies 2.92 Å from DD1 Glu16. This comparison uses a reference receptor placement, not the AFSample3 target-docking pose. It distinguishes useful geometric examples without assigning pH states or measuring masking. Supplementary Figure S1 locates these examples in the archived UMAP projection; Figure S2 and Table S7 show the actual cluster medoids. The selected examples are not the medoids, and whole clusters are not labeled masked or exposed.

2.4 AlphaProtein Novo and LigandForge expand the paired redesign assessment

AlphaProtein Novo supplied 1,000 sequences assessed against both EGFR species. Their mean human/mouse ipSAE ranged from 0 to 0.938258 (median 0.548874); the selected ten ranged from 0.929295 to 0.938258. Twenty DeltaForge affinity predictions ranged from 0.84 to 336.26 nM. LigandForge selection instead ranked all human and mouse complexes together, selecting nine sequences through their human scores and one through its mouse score. For those ten sequences, human ipSAE ranged from 0.921581 to 0.943554 and mouse ipSAE from 0.483749 to 0.923920. Twenty predicted dissociation constants (K_d) ranged from 18.21 to 450.24 nM. Smaller K_d values indicate stronger predicted binding. Different generation objectives and selection rules prevent a controlled comparison of the two design methods. Supplementary Figures S3–S4 show the archived ipSAE distributions for the full initial 1,000-design cohorts.

The LigandForge complexes also illustrate why overall shape and binding contacts should be checked separately. Human/mouse DockQ, which compares the binding interfaces, ranged from 0.0671 to 0.7736 even though whole-complex TM-scores ranged from 0.8095 to 0.9887. Similar overall structures did not guarantee similar interfaces.

Proton-PottsMPNN generated 1,000 variants for each of ten starting designs from each platform: 10,000 redesigns per platform and 20,000 overall [14, 15, 16, 17, 18]. Boltz-2 yielded 10,000 complete human/mouse pairs for AlphaProtein Novo and 9,450 for LigandForge [8]. Ten redesigns per parent were selected by mean human/mouse ipSAE, yielding 100 candidates per platform. Supplementary Figure S5 shows the LigandForge distributions: 18,900 predictions, including 450 pairs for parent 774 and 1,000 for each other parent. Supplementary Figure S6 combines the initial 1,000-design Novo distribution with all ten 1,000-redesign libraries and their paired species distributions. Each selected panel has 400 unforced JustHISpKa receptor-site estimates and 200 DeltaForge $K_d/\Delta G$ predictions [20, 17]. All 200 redesign sequences contribute to every panel of Figure 3.

Figure 2 summarizes generation, redesign and selection separately for the two platforms. The original designs, paired human/mouse structure predictions, pKa estimates and DeltaForge results provide the comparison group for these Proton-PottsMPNN redesigns. Parent-linked records distinguish sequence-redesign effects within a design route. Differences in original generation objectives, selection rules and historical chemical preparation limit comparisons between routes to descriptive assessments.

All ten AlphaProtein Novo starting binders had 1,000 complete redesign pairs in the final job-3835 assessment. Selecting the ten highest mean human/mouse ipSAE values within each parent yielded 100 distinct sequences, with mean scores from 0.930578 to 0.953662. JustHISpKa [20] supplied all 400 requested unforced EGFR-site estimates: 47 human and 45 mouse complexes had at least one estimate in 6.5–7.4. DeltaForge [17] supplied affinity and free-energy predictions for all 200 species-specific complexes, with species-specific K_d values from 0.61 to 1032.03 nM and ΔG values from -12.57 to -8.17 kcal mol⁻¹. These 100 candidates are represented in all five main panels; Supplementary section S19 preserves their identities and species-specific scores.

2.5 Expert redesigns show why pKa sensitivity needs to be checked

Fifteen additional rational single substitutions from Novo-0037 and Novo-0470 yielded 30 native human/mouse complexes. A separate D46 branch tested V8G and V8S in its DD2 extension, with five models per species and an unchanged DD1 sequence; the WT control was refolded alongside them. All 60 collected models were scored by DeltaForge and JustHISpKa on the native bound proteins. Candidate figures use the highest-confidence model independently per species. All sample-specific predictions remain in the campaign packages. The pKa plots retain species-specific estimates, and no predictions were averaged across model samples. Figure 3C summarizes affinity by the arithmetic mean of the two species.

The 22 expert-designed mutants yielded 220 Boltz-2 models, of which 44 were selected by confidence. Their median DockQ against the original RFD3 binding pose (the relative arrangement of binder and receptor) was 0.0221, indicating poor agreement with that pose. Parent and mutant predictions used different structure-template and glycan settings, so the difference cannot be attributed to the sequence changes alone (Supplementary section S4). Supplementary Figure S11 compares species-specific confidence and predicted affinity changes for the 39 expert redesigns relative to their exact-sequence parents.

The charge assigned to nearby acidic residues also affected some histidine predictions. For mouse EGFR H100, JustHISpKa gave 6.771 or 4.716 with the DD1-062 A76Q+I88Q mutant, and 8.173 or 2.338 with D46, depending on whether nearby acidic residues were fixed as unprotonated or protonated. These are predictions for deliberately imposed protonation patterns, not measurements or uncertainty intervals. They motivate sensitivity checks but are distinct from the common PROPKA pKa shifts used in the Potts analysis.

2.6 Receptor histidine pKas identify potentially responsive binding sites

Figure 3D,E shows EGFR His37 and His100 pKa estimates for the same 394 candidates as panels A, B and C. Shaded bands show the inclusive 6.5–7.4 interval; the supplementary CSVs identify candidates with at least one receptor histidine in that interval. A site in this interval may change its protonation between the two conditions, making it relevant to the design hypothesis. It does not establish whether binding becomes stronger or weaker. These receptor-site predictions and panel B answer different questions: panel B varies selected binder residues while holding receptor protonation fixed.

The DD1-062 A76Q+I88Q mutant and D46 fusion are now included in every main panel. JustHISpKa could not supply a unique unforced mouse His100 estimate because nearby acidic groups also titrate. Panel E therefore shows both retained acid-state predictions for each construct, while their unforced values remain missing in the data files. For the mutant, fixing binder Asp55 as deprotonated or protonated gives 6.771 or 4.716; for D46, fixing Glu16/Asp60 gives 8.173 or 2.338. Neither scenario is assigned to pH 6.5 or 7.4, and the values are not averaged. Across the 394 candidates, bound protein-complex JustHISpKa [20] supplies 394 human and 392 mouse His37/His100 pairs; 392 candidates have numeric pairs in both species. Unforced retries reproduced both mouse His100 failures. These structure-based estimates do not establish calibrated binder-induced pKa shifts.

Supplementary Figures S9–S10 retain the complete 394-candidate confidence and receptor-pKa maps; section S21 links the shared main-figure cohort to the source records and explicitly conditional outputs. The original, redesign and Foundry supplementary site tables preserve exact values, identities and prediction scope.

2.7 Mean predicted affinity across species

Figure 3C compares mean human/mouse interface confidence with mean predicted affinity for the same 394 sequences as all other panels. Both means are arithmetic means of the two species values; lower predicted K_d indicates stronger predicted binding. A mean can hide a weaker prediction for one species, so Supplementary Figure S8 separates the species for this same cohort. The comparison includes 100 Foundry designs, 100 LigandForge-derived redesigns and 100 Novo-derived redesigns with verified human and mouse affinity predictions. Historical preparation and scoring differences support descriptive comparison rather than a controlled comparison of generators. DeltaForge [17] accepted no pH input, so these affinity predictions do not measure pH selectivity.

2.8 Complete-data ranking distinguishes current priorities from the historical submission

The competition submission preceded completion of the full assessment. We preserve the earlier frozen twenty-sequence selection snapshot separately and apply its recovered selection rule retrospectively to the current 394 exact sequences shared by Figure 3. The historical workflow ranked nineteen candidates and reserved one slot for the exact Figure 1 D46 fusion. The updated ordering evaluates all available candidates without reserving a comparator slot or replacing the submitted sequences. Thirty Novo-derived redesigns have been added since the retained 364-candidate selection snapshot, and native bound-complex JustHISpKa estimates now replace the earlier original-LigandForge PROPKA and DD1-parent receptor-only annotations [17, 20]. Updated pKa contexts affect thirteen sequences per species. The shared 364 candidates retain the same species-specific ipSAE and K_d predictions.

We required four numeric, unforced EGFR D3 His37/His100 pKa estimates and positive nominal Potts differences, $S(7.4) - S(6.5)$, in both species. These conditions admit 103 candidates; 289 additional candidates have complete receptor pKas but fail the two-species Potts gate, and two lack an unforced mouse His100 estimate. Conditional values were not substituted. Five eligible candidates place all four pKas within the inclusive 6.5–7.4 interval and form the first tier. Within this tier, we rank by the equal mean of two empirical midrank percentiles over the 103 eligible candidates: higher minimum human/mouse ipSAE and lower maximum human/mouse predicted K_d . Each term thus reflects the weaker species. For each metric, the percentile is the number of eligible candidates with a less favorable value plus half the number of other candidates tied at that value, divided by 102. DeltaForge [17] ΔG predictions receive no additional weight alongside the related K_d predictions.

The remaining 98 eligible candidates follow in a second tier. Each receptor pKa contributes its distance to the interval, with zero distance inside it. We order candidates by the largest of the four distances, then their summed distance, then the same confidence/affinity quality score. Remaining quality ties favor a positive Potts direction in both species at all common binder-center pKa shifts of -1 , 0 and $+1$, followed by lower worse-species K_d , higher weaker-species ipSAE and exact candidate identifier. All comparisons use unrounded values; no additional sequence-diversity constraint is imposed. The updated eligible top twenty share seventeen sequences with the earlier frozen panel, and eighteen retain a positive Potts direction at all tested shifts. Original LigandForge-559 and Foundry-126/model-4 do not meet that stability criterion. D46 V8G retains a mouse ipSAE of 0.184 despite its comparatively close receptor pKas, illustrating the consequence of prioritizing pKa proximity before confidence in the second tier.

Supplementary Table S15 and Supplementary Data 5 retain all 394 assessments, the 103 eligible ranks and the current top twenty. Excluded candidates are listed without a rank in source-inventory order; that display order carries no prioritization. The D46 fusion in the earlier frozen panel was a mandatory comparator exception: its 91-residue sequence contains DD2 residues 1–14 followed by DD1 residues 15–91, without an added tag or linker. Its conditional mouse His100 prediction and negative nominal mouse Potts difference exclude it from the eligible current ranking. The retrospective ordering preserves that distinction and does not constitute a new competition submission or evidence of measured pH selectivity.

3 Discussion

3.1 Binding contacts and masks require different evidence

The model in Equation 1 separates the strength of an exposed binding surface from the fraction of molecules that expose it. For the binding-contact aim, three variants—062/0955, 062/0629 and 062/0436—had lower average Potts scores at pH 6.5 than at 7.4 in both species under every tested common pKa shift. Figure 3B uses this direction of change to prioritize experiments. It does not demonstrate a binding preference. Reweighting the original scores aligns the comparison with the intended 6.5/7.4 window, but the independent-site weighting remains an approximation.

Variant 195/0591 ranked higher by structural confidence yet favored the opposite pH in the Potts calculation. In this workflow, we first selected plausible interfaces and then evaluated their pH-dependent scores. Jacobsen and

colleagues instead used protonation-aware model objectives to select their redesigned experimental library [15]. The disagreement between our two rankings shows why the choice of selection rule matters; it does not establish a defect in Proton-PottsMPNN.

3.2 Preserving the binding fold does not establish masking

D46 retained the DD1 fold, showing that adding this short extension need not destroy the binding module. However, lysine–histidine–lysine sequence order did not produce the intended contacts in the reviewed structures. The additional sampled structures include DD2–DD1 carboxylate proximities and different degrees of coverage of the reference binding face. These geometric observations are distinct from the intended lysine network and do not demonstrate a pH transition. Future masks require both suitable contacts and surface coverage, followed by evidence that these change with pH. D46 does not establish functional states.

3.3 Interpretation across design routes

The current top twenty in Results provide a retrospective prioritization from the completed 394-candidate assessment. The earlier frozen panel preserves the selection made from the data available at that time. An earlier proposed test panel contained the three pKa-shift-stable variants and their starting binder, DD1-062. The DD1 portion of D46 without its extension would provide a control for the effect of adding DD2. Those proposed controls are separate from the historical submission and have not been experimentally validated. The primary human measurement is $K_d(7.4)/K_d(6.5)$ on the full extracellular receptor; a ratio above one means stronger binding at pH 6.5. Testing mouse EGFR only at pH 6.5 can assess recognition of the second species but cannot establish mouse pH selectivity. A future mask that passes the contact-and-coverage rule should be compared with its unextended DD1 and a control that disrupts masking.

The completed AlphaProtein Novo and LigandForge expansions each compare ten selected variants from every starting design. Parent-linked sequences and scores preserve those within-route comparisons [16, 17]. The 100 redesigns per platform contribute to all five main assessments, including paired affinity and receptor-site pKas. Original selection rules, the 450-pair LigandForge parent-774 library and pH-independent DeltaForge predictions limit cross-route interpretation. The results do not establish which generator performs best under matched experimental conditions.

Most calculations used D3, whereas the intended assays use human Met1–Ser645 and mouse Met1–Ser647 extracellular receptors. Neighboring domains, receptor motion, sugar modifications and how the receptor is held in the assay may affect binding [27]. The Potts weighting does not include coupled changes in protonation or alternative structures of the free proteins. DeltaForge accepted no pH input, and AFSample3 cluster sizes are not equilibrium populations. These limitations constrain interpretation of the computational candidates and masking hypothesis.

4 Conclusions

This study separates two ways to seek stronger EGFR binding under mildly acidic conditions: improving binding contacts and exposing a masked binding surface. It identifies three DD1-062 variants whose model-score preference persists under the tested pKa shifts in both species. D46 preserves the binding fold and provides sampled geometries for assessing DD2 coverage, while pH-dependent masking remains untested. These results, together with the completed paired assessments of 100 AlphaProtein Novo and 100 LigandForge redesigns, provide a candidate test panel and a structural criterion for future masking designs.

5 Methods

5.1 Study records and reproducibility

Preparation, sequence generation, structure prediction and analysis were recorded separately. Each score was linked to an exact sequence, receptor species, chain assignment, residue-numbering map and chemical preparation. Changes to inputs or settings were saved as new runs, preserving the earlier records. Software and model versions, file checksums and unresolved gaps in older records are documented in the accompanying audit. Scoring used the original predicted coordinates; structures aligned for figures were saved separately. This manuscript revision did not generate new protein designs; the expanded Potts analysis rescored retained structures.

5.2 Retrospective candidate ranking

The current ranking reuses retained native-model predictions and joins candidate and species records by exact amino-acid sequence SHA-256. All 788 species-specific Potts records were verified against their native-model hashes and endpoint arithmetic. Before analyzing the expanded inventory, the successor helper reproduced the earlier six-strict-plus-thirteen-near ordering and its quality values from the frozen 364-candidate snapshot. The eligibility, pKa-distance and empirical-percentile rules are specified in Results and Supplementary section S23. Two unresolved unforced mouse His100 values remain blank. No new predictions or sequence generation were required for this ranking. The twenty highest eligible ranks define the current panel; the historical mandatory D46 comparator is not imposed on it. Supplementary Table S15 rounds values for display; the accompanying CSVs retain the unrounded inputs and ranking terms.

5.3 Shared structure-prediction and scoring protocol

Boltz-2 predicted binder structures and their complexes with human and mouse EGFR [8]. Most campaigns used one model per complex; expert-designed mutants used five, followed by confidence-based selection. Settings common to these calculations and campaign-specific differences are given in Supplementary section S9.

The assessment methods answer different questions. ipSAE estimates confidence in the predicted interface from alignment-error information [9]. TM-score and RMSD describe agreement with a reference structure, and DockQ describes agreement at the interface [29, 28]. DeltaForge predicts affinity [17]. Higher interface confidence does not itself mean stronger binding. The Potts difference is a separate sequence-structure score in model units, not a binding free-energy difference ($\Delta\Delta G$).

Equation 1 assumes that the mask equilibrates rapidly and only exposed DD1 can bind. A fuller model would need the possible structures and protonation patterns of both free and bound proteins, together with the effects of attaching the mask. Those quantities were not calculated here.

5.4 EGFR structures and residue numbering

Human EGFR coordinates came from structure 6ARU in the RCSB Protein Data Bank [21, 26], linked to UniProt P00533 [22]. The mouse starting structure was the predicted AlphaFold Database model AF-Q01279-F1 [23, 25], linked to UniProt Q01279 [24]. Each D3 fragment contained 192 residues. The project numbered both fragments 310–501, so local His37/His100 correspond to H346/H409 in that convention. The original structures differ: the human fragment is chain A 310–501, whereas the mouse model v6 fragment is chain A 334–525, with histidines at 370/433. Residues were matched by sequence, not by assuming that the two source structures shared numbering.

Calculations on proteins without glycans, follow-ups including glycans, and protein-only scoring copies are distinguished throughout the records.

5.5 Aspartate-directed DD1 design and DD2 extensions

RFdiffusion3 run through Foundry [7, 18] generated 100 placements of two aspartates near human EGFR His37/His100. Their identities were fixed but their coordinates could move (seed 37; 25 batches of four). Four placements were developed into 100 binders each, 40–100 residues long, while preserving the placed aspartate geometry and receptor coordinates (seeds 137–140). All 400 binders were predicted with both EGFR species. Selection required ipSAE > 0.63 and binder TM-score > 0.75 in each species, retaining two designs from placement pair 11. ipSAE used 15 Å predicted-alignment-error and distance cutoffs, taking the larger of the two chain-direction scores. Binder TM-score was calculated relative to the RFD3 design.

For DD2, Foundry RFD3 [7, 18] added 10–20 residues to the N-terminus of a fixed 77-residue DD1 (seed 139; 250 batches of four). The added sequence had to contain lysine, histidine and lysine in that order, with variable spacing and positions. DD1 E2/D46 were designated as contact sites, separate from the original DD1 D11/D13 aspartate pair. The 1,000 fusions were predicted alone and with human or mouse EGFR, producing 3,000 calculations. The five fusions with the highest confidence in their isolated fold (pTM) advanced to AFSample3 submission.

Global original-DD1 audit. We audited original Foundry-lineage RFdiffusion3 designs [7, 18] across the 10,000-sequence human-target discovery cohort, the 342-sequence original DD1 cohort and the 400-sequence aspartate-directed cohort. Eligibility required native Boltz-2 model_0 structures and PAE sidecars for both species. Exact full binder sequences defined uniqueness; none were duplicated within this 10,742-sequence union. All structures and sidecars matched retained hashes, binder sequences and species-specific receptor sequences. We ranked arithmetic mean human/mouse ipSAE using the larger chain-direction score with 15 Å PAE and distance cutoffs, and retained the fixed top 100 requested for the comparison. The cutoff was 0.8215955. The three manually selected redesign parents DD1-062, DD1-195 and DD1-310 had global ranks 326, 116 and 820, respectively, and were retained separately because their redesign work was already completed. This audit used historical scores and native-coordinate verification, with no new folding. Supplementary section S21 includes these 100 sequences in the full confidence and pKa maps. Their completed human and mouse affinity predictions now permit inclusion in all five main panels. Supplementary Figure S12 compares mouse affinity predictions for these designs and the 100 selected LigandForge-derived Proton-PottsMPNN redesigns [14, 15, 17, 18].

5.6 AlphaProtein Novo and LigandForge design and selection

AlphaProtein Novo [16] generated 1,000 sequences and structures of 80–120 residues around fixed human D3 (seeds 0–999; 1,000 diffusion steps). It used the whole D3 fragment as input, without an implemented attraction objective for named contact sites. No LigandMPNN resequencing or AlphaFold structure prediction was performed within this generation step. Separate Boltz-2 predictions produced 2,000 human/mouse complexes. Ten sequences were selected by mean human/mouse ipSAE, using their identifiers to resolve ties. Recalculated selected scores agreed to six decimal places. Later expert-design campaigns specified eight single substitutions for Novo-0037 and seven for Novo-0470; their paired predictions are included in the rational-redesign results.

LigandForge, accessed through the LigandAI platform [17], supplied 1,000 sequences of 20–60 residues, followed by 2,000 Boltz-2 complex predictions forming 1,000 matched human/mouse pairs. A downloaded generation-summary PDF documents contact-site labels H370/Q408/T430/H433. Sequence alignment maps them to local D3 H37/Q75/T97/H100, or project reference positions 346/384/406/409. The local positions are conserved in mouse EGFR. Supplementary Table S3 and section S8 preserve the mapping and generation identifier. The web interface displayed a human pH 6.5/7.4 objective, but the original submitted settings and model version have not been recovered. The documented objective therefore does not establish how pH was used by the generator. AlphaProtein Novo generation did not explicitly use pH. LigandForge selection ranked all species-specific complexes together, then included each selected sequence’s prediction with the other species. This differs from selection by the human/mouse mean.

Redesign from the selected starting binders. Each platform’s expansion generated 1,000 Proton-PottsMPNN variants for each of ten starting designs and assessed them with Boltz-2 [14, 15, 16, 17, 18, 8]. The LigandForge selection retained ten variants per parent by mean human/mouse ipSAE from 9,450 verified

completed pairs. The final AlphaProtein Novo job-3835 collection contained all 20,000 native complexes, yielding 10,000 complete human/mouse pairs and 1,000 pairs for each of ten parents. Ten Novo variants per parent were selected by descending arithmetic mean human/mouse ipSAE, with stable design identity resolving ties. These 100 distinct sequences extend the preceding 294-candidate comparison to 394. Exact sequences, parent relationships, source-model hashes and generation records were verified before score integration. The Novo native samples used Boltz-2 2.2.1, seed 0, one diffusion sample, three recycling steps and 200 sampling steps, with the retained receptor MSA, empty binder MSA and no templates. The larger chain-direction ipSAE v4 score used 15 Å PAE and distance thresholds [9]. Both redesign panels used standalone Proton-PottsMPNN; the Novo implementation was v6. Foundry [18] is related-framework attribution rather than the runtime used for these variants. JustHISpKa receptor pKa predictions and DeltaForge predicted $K_d/\Delta G$ values were subsequent assessments, independent of the ipSAE selection [20, 17].

5.7 Proton-PottsMPNN redesign and pH-score calculation

Proton-PottsMPNN [15] extends PottsMPNN [14] by representing neutral and charged forms of histidine, aspartate and glutamate separately. It produced 1,000 variants each of DD1-062, DD1-195 and DD1-310, optimizing groups of residues while keeping the starting backbone fixed (block descent). The model scores individual residues and residue pairs in their structural surroundings. The starting RFD3 designs came from Foundry [18]; redesign used the separate Proton-PottsMPNN v6 implementation and trained model.

Preparation used HBPLUS [13] to identify hydrogen-bond geometry. These features helped assign structural protonation labels for the model; they did not specify a numerical pH. The same preparation was used after Boltz-2 prediction. Supplementary section S9 gives the hydrogen-bond cutoffs, labeling details and model preparation.

For each selected DD1 histidine, aspartate or glutamate allowed to change protonation, PROPKA estimated a pKa from the binder-EGFR complex. The site identities matched the saved DD1 sequences in all 60 complexes. Receptor sites, including His37/His100, and other binder sites kept their configured protonation states. The Henderson-Hasselbalch relation converted pKa estimates to protonation probabilities at each pH. Assuming independent sites, we combined these probabilities to weight each complete pattern, then averaged its saved Potts score for the complex minus the two separate partners. PROPKA supplied the pKas; this averaging was a separate analysis of Potts scores. Supplementary section S1 gives the equations.

We subtracted the weighted score at pH 6.5 from that at 7.4. Positive differences favor pH 6.5 in the model; zero indicates no preference between the endpoints. The difference has no calibrated significance threshold or conversion to affinity. Coordinates, individual protonation-pattern scores and all non-varied states were fixed. Shifting every varied-site pKa by -1 , 0 and $+1$ tested sensitivity to the pKa assumptions. Before reweighting at 7.4, we reproduced the original 6.5/7.3 scores and shift differences exactly. No new Potts evaluation or pKa calculation was needed. No signs changed across 180 species/shift comparisons; the largest absolute change in the unshifted difference was 0.233 model units (Supplementary Table S5). This arithmetic average does not calculate a thermodynamic partition function or model protonation of the unbound proteins.

For the expanded five-panel comparison, an additional CPU-only assessment enumerated binder protonation patterns for all 394 candidate sequences on their retained human and mouse native complexes (788 complexes). It used the pinned standalone Proton-PottsMPNN implementation and the same structural feature preparation [13, 14, 15]; Foundry [18] is a related-framework attribution. Recorded design centers were retained for the 230 Proton-PottsMPNN redesigns. For the other 164 candidates, we chose up to two binder histidine, aspartate or glutamate side chains nearest the receptor in the human model and used those same sequence positions in mouse. This geometric rule was chosen before inspecting the scores; it does not establish that each selected site forms an interface contact. Receptor protonation and other binder states were fixed. Nonprotein atoms were omitted explicitly, without minimizing the protein coordinates. PROPKA supplied binder-center pKas for independent-site weighting at pH 6.5 and 7.4, repeated under common shifts of -1 , 0 and $+1$. The two archived control complexes reproduced their previous individual-pattern scores and pKas exactly. Supplementary section S21 records the selection rule, completion counts and exclusions. This new evaluation is distinct from the original 30-variant endpoint reweighting described above. The 60 complexes for the final three Novo parents used the same pinned scorer, checkpoint, recorded generation centers and validated CPU-only Slurm platform as the preceding 728 complexes; the same two archived controls were repeated without changing scientific settings.

5.8 Histidine pKa prediction and chemical preparation

JustHISpKa [20] predicted EGFR His37/His100 pKas in bound protein complexes and historical receptor-only controls. Controls used receptor coordinates extracted from the complex, rather than separately relaxed free receptors. The environment labeled AmberTools26 used PDB4Amber, LEaP and ambpdb [32], with ff14SB [33], to prepare atom connectivity, charges and hydrogen positions, including specified disulfide bonds. Non-hydrogen coordinates were kept fixed without energy minimization. NRC conversion scripts adjusted atom/residue names and bond types to the predictor’s MOL2 file format without changing connectivity or coordinates.

JustHISpKa samples histidine side-chain orientations and proton positions, calculates electrostatic interactions in a continuum-solvent model, and corrects for the amount of buried side-chain surface [20]. Hydrogens added during preparation do not establish the actual protonation state. Glycans were omitted from the expert-mutant pKa calculations. When a calculation without imposed neighboring-acid states did not give a unique answer, we separately fixed those acidic residues as protonated or unprotonated. The resulting conditional predictions were neither averaged nor treated as uncertainty intervals. Supplementary Materials give these values and the chemistry controls.

PROPKA provided binder-site pKas used in the Potts calculation; receptor-site estimates in Figure 3D,E now all use JustHISpKa [20]. Both predictors remain distinct from measured pKas. Figure 3D,E uses one unforced His37/His100 pair per candidate and species where available, with two explicit conditional points for each unresolved mouse His100 case. Either predicted pKa in the inclusive 6.5–7.4 interval qualifies; classification uses the values before rounding. Missing unforced values remain blank; the four conditional outputs are labeled separately, and human and mouse values were not averaged. The ten original LigandForge binders and three DD1 parents were rescored in their exact native bound protein complexes with JustHISpKa 1.0.2, AmberTools26/ff14SB and NRC correction; all 52 requested site predictions completed. Earlier PROPKA and receptor-only results are preserved. Novo-0293 uses specified C2–C84 and C10–C73 disulfides after sequence and coordinate checks; the alternative preparation is reported in Supplementary Table S1. The rational additions use new JustHISpKa 1.0.2 calculations on verified native bound proteins, with receptor B:37/B:100, four checked receptor disulfides, fixed protein heavy atoms and no minimization. Native D46 glycans were retained in structural files and omitted from protein-only scoring.

For the original Foundry top 100, H37/H100 calculations were attempted in all 200 native human/mouse complexes with JustHISpKa 1.0.2, without externally imposed protonation forces [20]. Atom, topology, local-to-global numbering and source-hash audits were retained. Four canonical receptor disulfides plus any unambiguous native binder SG pairs below 2.5 Å were assigned; additional binder bonds are explicitly inferred. Hydrogens and terminal OXT were added with Amber ff14SB, without heavy-atom minimization. Twenty-nine contexts from 17 binders have inferred binder bonds. Six designs have different inferred assignments across species; these chemistry assumptions are preserved in the source-linked table. Local H37/H100 correspond to human source H346/H409 and mouse source H370/H433; the shared project-reference numbering is H346/H409.

For the new LigandForge-derived panel, ten variants per parent were retained by descending arithmetic mean human/mouse ipSAE, then minimum species score and stable design identity. Only verified completed pairs were eligible: 9,450 of 10,000 pairs, with 450 for parent 774 and 1,000 for every other parent. JustHISpKa used native protein-only Boltz-2 complexes (binder A, receptor B), AmberTools 26 ff14SB and NRC correction, with fixed heavy atoms, no minimization and no forced site states [20]. All 400 requested receptor-site estimates were checked against raw outputs and H37/H100–H346/H409–global-site mappings. A linked successor preserved eight verified intra-binder disulfides in eight complexes across five designs, alongside the four receptor bonds; 192 unaffected records were reused unchanged. Bonds were species-specific. The short 1.589 Å binder bond in LF-559/0061 Mouse remains a geometry warning. No PROPKA or conditional estimates replaced requested values. Supplementary section S11 records preparation and deployed versions.

The 100 Novo-derived pairs were prepared from their native protein-only Boltz-2 complexes, with binder A and EGFR B. AmberTools ff14SB and the NRC conversion preserved source protein heavy-atom coordinates, with no minimization [20, 32, 33]. Every complex retained the four native EGFR disulfides and the exact unambiguous intra-binder bonds detected by the validated Amber preparation; absent or geometrically unrecognized binder bonds were not enforced from the parent’s pattern. Sequence identities, chain roles, histidine mappings and final heavy atoms were audited for all 200 complexes. JustHISpKa 1.0.2 supplied all 400 B:37/B:100 estimates, corresponding to project full-EGFR H346/H409, without imposed neighboring-acid protonation states. Conditional

predictions and PROPKA were not used as substitutes. Native source complexes contained no glycans.

5.9 Expert-guided sequence changes

Researchers specified 37 exact DD1 redesign sequences: the earlier 22 variants from DD1-062, DD1-195 and DD1-310, and 15 additional single substitutions from Novo-0037 and Novo-0470 (Supplementary sections S3 and S13). Two further V8 substitutions alter DD2 within the D46 fusion while preserving its DD1 sequence. The native models, both-species affinity and receptor-site predictions for the new batches are retained in Supplementary Table S8. The following five-sample protocol describes the earlier 22 DD1 variants; the new Novo variants have one sample per species and the D46 WT/V8G/V8S batch has five samples per species. Each of the earlier 22 variants was predicted with both species using five Boltz-2 models per complex, retaining the specified glycans and receptor sequence-alignment/template choices. The highest model-confidence score selected one structure per complex, with sample number resolving ties. Atom connectivity, coordinates and the dimensions of the confidence and alignment-error arrays were checked before scoring. The 44 selected models were compared with the original RFD3 interfaces and unmutated Boltz-2 predictions for the same species. Differences in earlier template and glycan settings prevent attributing all structural changes to mutations alone.

5.10 Interface confidence, predicted affinity and structural agreement

ipSAE [9] uses predicted alignment errors for pairs of residues in different protein chains, taking account of how many residues meet the scoring criteria. The chain-direction and ranking rules differ by campaign as described above; software versions and parameters are documented in the audit.

DeltaForge, accessed through LigandAI [17], assessed 60 Proton-PottsMPNN complexes, 20 AlphaProtein Novo complexes, 20 LigandForge complexes, six starting-DD1 complexes and 44 earlier expert-mutant complexes. The new rational batches added 30 Novo complexes and 30 D46 WT/V8G/V8S complexes, including every native D46 sample. These new requests returned served K_d and ΔG predictions with scorer and calibration versions recorded per response. Figure 3 uses confidence-selected native samples, retaining the historical D46 control context as a separate comparison. Structure-file requests used receptor chain B and binder chain A. Expert mutants supplied protein-only copies with original protein coordinates. Four D46 fusion/parent requests instead supplied sequences and allowed DeltaForge to predict its own structures, using receptor A and binder B. All 100 Foundry candidates now have human and mouse DeltaForge predictions: a completed successor added 100 human native protein-complex requests and paired them with the retained mouse results. Receptor chain B and binder chain A were preserved, and all served values were verified against the responses. The completed LigandForge-derived redesign panel adds 200 native protein-complex requests, with receptor B and binder A, yielding 200 finite served ΔG values and positive K_d values. The deployed software versions are recorded separately from the cited preprint. The service exposed no explicit pH input.

The 200 Novo-derived DeltaForge structure requests via LigandAI [17] used receptor B and binder A. All returned finite served ΔG values and positive K_d values. Exact native protein sequences and source-coordinate hashes linked each response to its species-specific complex. Deployed scorer and calibration identifiers were retained per response, separately from the cited preprint. Both served quantities were used directly; K_d was not derived from ΔG .

US-align [29] calculated TM-score and RMSD relative to reference structures. DockQ v2 [28] compared interfaces after matching chains and residues. AlphaProtein Novo models were compared with their human design structures, including the equivalent mouse comparisons. Expert mutants were compared with the RFD3 designs and same-species starting-binder predictions, using `AB:AB` chain mapping and `no_align=True`. LigandForge human/mouse comparisons allowed 24 receptor sequence mismatches and averaged scores calculated in both directions. All reference interfaces were computational predictions; protein-only comparisons excluded glycan contacts.

5.11 AFSample3 structure sampling and grouping

AFSample3 [11], based on AlphaFold 3 [12], generated many possible D46 structures with human D3 and four attached glycans (Supplementary section S2). It used 200 random seeds and five samples per seed, giving 1,000 predictions, without an explicit pH input. Detailed settings are given in Supplementary section S9.

Structures were compared using distances between corresponding backbone C α atoms in the fusion. UMAP [31] provided a two-dimensional display in which structurally similar models could be grouped. The saved analysis chose four clusters using a separation criterion called the silhouette score (Supplementary Figure S1). The analysis contains one additional run-level structure, for 1,001 entries overall. Neither masking contacts nor coverage of the DD1 surface was used to form or label the original clusters. The later display selection measured DD2 proximity to a reference binding face after DD1 alignment, as detailed in Supplementary section S10. Figure 1 examples and the cluster medoids are located in Supplementary Figure S1; medoid structures are shown in Figure S2. These geometric comparisons do not assign pH states or equilibrium populations. The available ranking file covered only five samples from seed 200, not the whole set. Technical details and its confidence values are preserved in Supplementary section S9.

5.12 Summary statistics

For the paired affinity plots, both species had to have finite positive predicted K_d values and recorded ipSAE values for the same binder sequence. Human and mouse values were averaged arithmetically, before plotting K_d on a logarithmic axis. These means are summaries of two predictions, not an affinity for a mixture of receptors. Missing human predictions were not replaced by mouse values.

Counts, ranges and medians describe the specified sequences, complexes or predicted structures. Models for two species and repeated computational samples are not independent biological experiments. We did not derive statistical significance or binding uncertainty intervals from prediction scores. The pKa-shift analysis describes sensitivity to chosen assumptions, rather than a calibrated margin of error.

Acknowledgments

We thank Y.J. Huang, T. Acton, S. Zakian, and T.A. Ramelot for helpful discussions, and S. Collen for computational systems support.

Data and code availability

The Supplementary Materials provide the assessed fusion and human EGFR fragment sequences, expert-designed substitutions, candidate histidine pKa values, comparisons of the two pH endpoints and the AFSample3 structural clusters. Candidate sequences, original inputs and outputs, scoring tables and analysis code are saved in the campaign archive with records linking them to their sources and software settings. Supplementary Data 3A–3E provide one CSV per Figure 3 panel, containing stable sequence identifiers, exact names and sequences, documented campaign origins where recovered, and plotted metrics. All five files represent the same 394 sequences. Data 3E contains 396 uniquely identified points because two candidates have two explicit acid-state predictions; their missing unforced mouse His100 values remain blank. The files also record the pKa-window and acid-favoring sensitivity flags. Supplementary Data 4 provide all 394 candidate sequences, both pH endpoints for all 788 human/mouse complexes, their microstate scores and sensitivity results. The supplementary source tables retain all 10,000 Novo redesign pairs, parent-specific distributions and the selected 100 paired affinity/pKa assessments. Supplementary Data 5 provide the complete 394-candidate assessment, the 103 eligible ranks and the current top twenty, including exact sequences, unrounded ranking terms and native-model hashes. The manuscript PDFs and selected-model display gallery are available in the [project GitHub repository](#); bulk original structures, raw model outputs and private run records remain in the campaign archive.

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